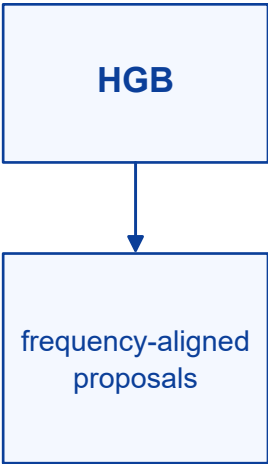


Dual-mode provenance triage for exploratory replacement proposals

A. Dual-mode proposal routes

Conservative Mode

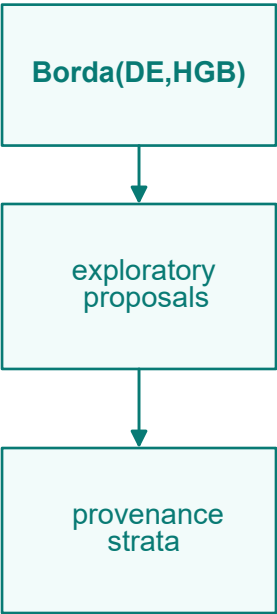
HGB route



Driven by HGB

Exploration Mode

Borda(DE,HGB) route



Driven by Borda(DE,HGB)

B. Provenance strata

G4 shared

$$K_{\text{HGB}} \cap K_{\text{Borda}}$$

consensus

G3 DE-elevated

$$K_{\text{Borda}} \setminus K_{\text{HGB}}$$

similarity-supported expansion

G2 Borda-emergent

$$K_{\text{Borda}} \setminus (K_{\text{HGB}} \cup K_{\text{DE}})$$

highest novelty

C. Coverage-limited A4C triage

Group	Coverage	Alert signal
G2	100%	46.85%
G3	100%	9.67%
G4	5.63% covered unknown	17.60% among covered; 94.37% unknown

A4C = computational triage only;
not experimental validation.